DIVYANSH NEGI

Backend & Systems Engineer · Distributed Architectures & Query Engines

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EDUCATION

Birla Institute of Applied Sciences (UTU) — *B.Tech in Computer Science and Engineering* **Bhimtal, India | Aug 2024 – May 2028**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, Compiler Design, Computer Networks, Theory of Computation.

TECHNICAL SKILLS

Languages:	Python, TypeScript, SQL, C++, C, JavaScript, Bash
Backend & Systems:	FastAPI, Node.js, Express, RESTful APIs, Apollo GraphQL, Server-Sent Events (SSE), Linux/Unix, AsyncIO
Databases & Storage:	PostgreSQL, ClickHouse Cloud, DuckDB (In-Memory Columnar OLAP), Redis, Schema Introspection
DevOps & Testing:	Docker, Git, GitHub Actions (CI/CD), Pytest, Model Context Protocol (FastMCP), Pydantic v2, PyMuPDF

TECHNICAL PROJECTS

OmniQuery AI — *Analytical Query Gateway & Dual-Engine Middleware* [GitHub | Live Demo] **Python, FastAPI, ClickHouse, DuckDB, FastMCP, Docker**

- Architected analytical query gateway with dual-engine failover: routes queries to ClickHouse Cloud OLAP, seamlessly falling back to in-memory DuckDB upon cluster degradation or network timeouts.
- Standardized runtime schema discovery via Model Context Protocol (FastMCP) over `information_schema`, cutting metadata token overhead by 40%.
- Engineered self-healing SQL reflection loop intercepting syntax exceptions to auto-repair queries, maintaining 99%+ execution success rate across 50,000 records.

Open-Source Support Assistant — *Agentic RAG & Retrieval Benchmarking* [GitHub] **Python, Groq (Llama 3.3 70B), SQLite FTS5, PostgreSQL, Grafana, Docker Compose**

- Built autonomous ReAct agent indexing ~1,400 chunked FastAPI docs and closed GitHub issues with persistent SQLite FTS5 search and self-correcting tool call recovery.
- Designed offline evaluation pipeline measuring 90.00% Hit Rate (top-5) and 0.7542 MRR across synthetic ground-truth test sets with LLM-as-a-judge grading.
- Containerized full observability stack via Docker Compose, logging latency, token consumption, and user feedback into PostgreSQL with real-time Grafana dashboards.

BidSahayak — *Resilient Procurement Document Ingestion Pipeline* [GitHub] **Python, FastAPI, Pydantic v2, PyMuPDF, Docker, Pytest**

- Developed automated document ingestion service in FastAPI, parsing multi-page municipal procurement PDFs with deterministic schema enforcement and structured validation.
- Constructed dual-layer extraction engine pairing PyMuPDF digital text parsing with deterministic regex fallback verification, eliminating hallucinated financial metadata and deadlines.
- Containerized service with multi-stage Dockerfile and deployed on Linux with structured health checks, achieving zero crashes across unstructured municipal specifications during hackathon benchmarks.

Turtleneck — *Developer Tooling & Deterministic Verification Harness* [GitHub | v1.0.0 Release] **Python, Pytest, Shell, GitHub Actions CI, WCAG 2.2**

- Built zero-dependency Python CLI developer tool enforcing layout archetypes and WCAG 2.2 AA accessibility invariants across AI-generated frontend codebases.
- Authored 74 hermetic unit and integration tests in Pytest validating color contrast math, keyboard focus indicators, and AST rule parsing with 100% CI pass rate.
- Engineered idempotent multi-target installer automating configuration injections across 5 development environments (Claude Code, Cursor, Windsurf, Copilot, Antigravity) with dry-run safety gates.

TECHNICAL TRAINING & CERTIFICATIONS

- **Full Stack Open (University of Helsinki)** — Verified Certificates in **Containers & Docker** (Sep 2026), **TypeScript** (Aug 2026), **GraphQL** (Jul 2026), and **Core Architecture** (Jun 2026).
- **LLM Zoomcamp 2026 (DataTalks.Club)** — Production LLM engineering, vector search, and evaluation harnesses (Credential ID: `6F7D81`).
- **Applied Data Science Lab (WorldQuant University)** — End-to-end data pipelines and statistical modeling (Apr 2026).
- **Oracle Cloud Infrastructure (OCI)** — AI Foundations Associate Certified (Oct 2025).